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Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2020

Subject: Analog and Digital Communication System

Time: 03 Hours

Code: 110501

Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven of the following questions: **[14]**

(a) What is the criterion for Carson's bandwidth calculation?

(b) List the advantages of DSB-FC modulation scheme.

(c) A signal has frequency component from 300 Hz to 1.8 kHz. What is the minimum possible rate at which the signal has to be sampled?

(d) Define the numerical aperture of step index fiber.

(e) What is the need of cladding?

(f) What are the advantages of geostationary satellite?

(g) Mention the uses of frequency discriminator in FM demodulation.

(h) Write the advantages of hand-off process.

(i) What do you mean by baseband signal?

(j) Compare the spectrum occupancy of PSK and FSK schemes.

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Q.2 Solve both questions : **[14]**

(a) What is the basic limitation of SSB modulation scheme? How is it eliminated by VSB modulation?

(b) Draw the block diagram of super heterodyne AM receiver and explain the function of IF amplifier.

Q.3 Solve both questions : **[14]**

(a) 80 micrometer core diameter single-mode fiber has a core refractive index of 2, relative refractive index difference of 0.3% and operating wavelength of 1.55 micrometer. Determine critical radius of curvature. Explain bending losses.

(b) Discuss the Armstrong's method of indirect FM generation in detail.

Q.4 Solve both questions : **[14]**

(a) What is ISI? Explain how ISI can be removed by employing a raised cosine filter.

(b) Derive the output SNR of AM envelop detector. Also calculate its figure of merit.

Q.5 Solve both questions : **[14]**

(a) Explain the convolution codes, their advantages and disadvantages.

(b) A sinusoidal message signal of peak voltage 20 V and peak frequency of 5 kHz is transmitted through 256 levels PCM system. The sampling rate is 25% higher than Nyquist rate Calculate the sampling frequency, bit rate, bandwidth, step size, maximum quantization error and SNR in dB.

Q.6 Solve both questions : **[14]**

(a) Explain the sampling process of a signal mathematically. Write how to generate PPM from PWM signal?

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(b) Derive the power calculation of AM signal.

Q.7 Solve both questions : **[14]**

(a) What is the probability of error in presence of white noise for BPSK system? We are required to transmit 2.08×10^6 binary digits per second with $P_b$le10^{-7}. The channel noise PSD is $Sn(f)=10^{-8}$. Determine the transmission bandwidth and signal power required in binary.

(b) What are the noise considerations in PCM? Explain the TDM-PAM concept.

Q.8 Solve both questions : **[14]**

(a) An angle modulated signal with carrier frequency $\omega_c = 2\pi\times10^5$rad/sec is given as follow: $\varphi_{FM}(t) = 10\cos(\omega_c t + 5\sin(3000t) + 10\sin(2000t))$. (i) Find power of modulated signal. (ii) Find the maximum frequency deviation. (iii) Find the maximum phase deviation. (iv) Estimate the bandwidth of FM signal.

(b) Explain the different types of diversity technique used in wireless communication system.

Q.9 Solve both questions : **[14]**

(a) Explain telemetry, tracking and command control system in detail.

(b) Explain Snell's law. What is total internal reflection? Explain with suitable diagram.

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